

SIMATS ENGINEERING ASSESSMENT TOOLS

COURSE NAME: Theory of Computation

COURSE CODE: CSA1304

COURSE OUTCOME COVERED

CO3: Construct regular expressions, and use the pumping lemma and closure properties to analyze regular languages. (BL:3)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:100

Duration : 1 Hour

Experiments

Code of Conduct:

*I **J.Goutham Kumar Reddy / 192373041** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

J.Goutham Kumar Reddy

SIMATS ENGINEERING ASSESSMENT TOOLS

Sl. No. Lab Experiments - Questions

- 1 Write a C program to check whether a given string belongs to the language defined by a Context Free Grammar (CFG)
 $S \rightarrow 0A1 \quad A \rightarrow 0A \mid 1A \mid \epsilon$
- 2 Write a C program to check whether a given string belongs to the language defined by a Context Free Grammar (CFG)
 $S \rightarrow 0S0 \mid 1S1 \mid 0 \mid 1 \mid \epsilon$
- 3 Write a C program to check whether a given string belongs to the language defined by a Context Free Grammar (CFG)
 $S \rightarrow 0S0 \mid A \quad A \rightarrow 1A \mid \epsilon$
- 4 Write a C program to check whether a given string belongs to the language defined by a Context Free Grammar (CFG)
 $S \rightarrow 0S1 \mid \epsilon$
- 5 Write a C program to check whether a given string belongs to the language defined by a Context Free Grammar (CFG)
 $S \rightarrow A101A \quad A \rightarrow 0A \mid 1A \mid \epsilon$

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

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ANSWERS

1. CFG: S to 0A1, quad A to 0A mid 1A mid epsilon

Language Definition: Generates all binary strings of length ≥ 2 that start with 0 and end with 1.

C Program:

```
#include <stdio.h>

#include <string.h>

#include <stdbool.h>

bool is_member_g1(const char *str) {

    int len = strlen(str);

    if (len < 2) return false;

    if (str[0] != '0' || str[len - 1] != '1') return false;

    for (int i = 0; i < len; i++) {

        if (str[i] != '0' && str[i] != '1') return false;

    }

    return true;

}

int main() {

    const char *test_cases[] = {"01", "001", "0101", "0111", "101", "000", "0", ""};

    int num_tests = sizeof(test_cases) / sizeof(test_cases[0]);

    printf("Grammar 1: S -> 0A1, A -> 0A | 1A | epsilon\n");

    printf("-----\n");

    for (int i = 0; i < num_tests; i++) {

        const char *str = test_cases[i];

        printf("Input: \"%s\" \t-> %s\n", str, is_member_g1(str) ? "ACCEPTED" : "REJECTED");

    }

}
```

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```
    return 0;
}
```

Output:

Grammar 1: S -> 0A1, A -> 0A | 1A | epsilon

```
-----
Input: "01"  -> ACCEPTED
Input: "001" -> ACCEPTED
Input: "0101" -> ACCEPTED
Input: "0111" -> ACCEPTED
Input: "101"  -> REJECTED
Input: "000"  -> REJECTED
Input: "0"    -> REJECTED
Input: ""     -> REJECTED
```

2. CFG: S to 0S0 mid 1S1 mid 0 mid 1 mid epsilon

Language Definition: Generates all binary palindromes (including odd-length, even-length, and empty strings).

C Program:

```
#include <stdio.h>

#include <string.h>

#include <stdbool.h>

bool is_member_g2(const char *str) {

    int len = strlen(str);

    for (int i = 0; i < len; i++) {

        if (str[i] != '0' && str[i] != '1') return false;

    }

    int start = 0, end = len - 1;

    while (start < end) {

        if (str[start] != str[end]) return false;

        start++;
```

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```
        end--;

    }

    return true;
}

int main() {

    const char *test_cases[] = {"", "0", "1", "00", "11", "010", "1001", "011010", "1010"};

    int num_tests = sizeof(test_cases) / sizeof(test_cases[0]);

    printf("Grammar 2: S -> 0S0 | 1S1 | 0 | 1 | epsilon\n");

    printf("-----\n");

    for (int i = 0; i < num_tests; i++) {

        const char *str = test_cases[i];

        printf("Input: \"%s\" \t-> %s\n", str, is_member_g2(str) ? "ACCEPTED" : "REJECTED");

    }

    return 0;
}
```

Output:

Grammar 2: S -> 0S0 | 1S1 | 0 | 1 | epsilon

Input: "" -> ACCEPTED

Input: "0" -> ACCEPTED

Input: "1" -> ACCEPTED

Input: "00" -> ACCEPTED

Input: "11" -> ACCEPTED

Input: "010" -> ACCEPTED

Input: "1001" -> ACCEPTED

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Input: "011010" -> REJECTED

Input: "1010" -> REJECTED

3. CFG: S to 0S0 mid A, quad A to 1A mid epsilon

Language Definition: Generates strings of the form $0^n 1^m 0^n$ where $n \geq 0$, $m \geq 0$.

C Program:

```
#include <stdio.h>

#include <string.h>

#include <stdbool.h>

bool is_member_g3(const char *str) {

    int len = strlen(str);

    int i = 0, j = len - 1;

    for (int k = 0; k < len; k++) {

        if (str[k] != '0' && str[k] != '1') return false;

    }

    while (i < j && str[i] == '0' && str[j] == '0') {

        i++;

        j--;

    }

    for (int k = i; k <= j; k++) {

        if (str[k] != '1') return false;

    }

    return true;

}
```

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```
int main() {  
  
    const char *test_cases[] = {"", "111", "00", "010", "0011100", "000", "01100", "0100"};  
  
    int num_tests = sizeof(test_cases) / sizeof(test_cases[0]);  
  
    printf("Grammar 3: S -> 0S0 | A, A -> 1A | epsilon\n");  
  
    printf("-----\n");  
  
    for (int i = 0; i < num_tests; i++) {  
  
        const char *str = test_cases[i];  
  
        printf("Input: \"%s\" \t-> %s\n", str, is_member_g3(str) ? "ACCEPTED" : "REJECTED");  
  
    }  
  
    return 0;  
  
}
```

Output:

Grammar 3: S -> 0S0 | A, A -> 1A | epsilon

Input: "" -> ACCEPTED

Input: "111" -> ACCEPTED

Input: "00" -> ACCEPTED

Input: "010" -> ACCEPTED

Input: "0011100" -> ACCEPTED

Input: "000" -> REJECTED

Input: "01100" -> REJECTED

Input: "0100" -> REJECTED

4. CFG: S to 0S1 mid epsilon

Language Definition: Generates strings of the form $0^n 1^n$ where $n \geq 0$.

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C Program:

```
#include <stdio.h>

#include <string.h>

#include <stdbool.h>

bool is_member_g4(const char *str) {

    int len = strlen(str);

    if (len % 2 != 0) return false;

    int n = len / 2;

    for (int i = 0; i < n; i++) {

        if (str[i] != '0') return false;

    }

    for (int i = n; i < len; i++) {

        if (str[i] != '1') return false;

    }

    return true;

}

int main() {

    const char *test_cases[] = {"", "01", "0011", "000111", "0101", "001", "00011", "10"};

    int num_tests = sizeof(test_cases) / sizeof(test_cases[0]);

    printf("Grammar 4: S -> 0S1 | epsilon\n");

    printf("-----\n");

    for (int i = 0; i < num_tests; i++) {

        const char *str = test_cases[i];

        printf("Input: \"%s\" \t-> %s\n", str, is_member_g4(str) ? "ACCEPTED" : "REJECTED");

    }

    return 0;

}
```


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Output:

Grammar 4: $S \rightarrow 0S1 \mid \epsilon$

Input: "" -> ACCEPTED

Input: "01" -> ACCEPTED

Input: "0011" -> ACCEPTED

Input: "000111" -> ACCEPTED

Input: "0101" -> REJECTED

Input: "001" -> REJECTED

Input: "00011" -> REJECTED

Input: "10" -> REJECTED

5. CFG: $S \rightarrow A101A$, quad $A \rightarrow 0A \mid 1A \mid \epsilon$

Language Definition: Generates any binary string containing "101" as a substring.

C Program:

```
#include <stdio.h>
#include <string.h>
#include <stdbool.h>
```

```
bool is_member_g5(const char *str) {
    int len = strlen(str);
    for (int i = 0; i < len; i++) {
        if (str[i] != '0' && str[i] != '1') return false;
    }

    return strstr(str, "101") != NULL;
}
```

```
int main() {
    const char *test_cases[] = {"101", "01010", "1110100", "00101", "100", "010", "111", ""};
    int num_tests = sizeof(test_cases) / sizeof(test_cases[0]);

    printf("Grammar 5:  $S \rightarrow A101A$ ,  $A \rightarrow 0A \mid 1A \mid \epsilon$ \n");
    printf("-----\n");
    for (int i = 0; i < num_tests; i++) {
        const char *str = test_cases[i];
```

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```
    printf("Input: \"%s\" \t-> %s\n", str, is_member_g5(str) ? "ACCEPTED" : "REJECTED");  
}  
return 0;  
}
```

Output:

Grammar 5: S -> A101A, A -> 0A | 1A | epsilon

Input: "101" -> ACCEPTED

Input: "01010" -> ACCEPTED

Input: "1110100" -> ACCEPTED

Input: "00101" -> ACCEPTED

Input: "100" -> REJECTED

Input: "010" -> REJECTED

Input: "111" -> REJECTED

Input: "" -> REJECTED